

A COMPARISON OF THE CLASSIFICATION PERFORMANCE OF TWO APPROACHES TO POLYCHOTOMOUS LOGISTIC REGRESSION

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Summary: Logistic regression is frequently used to classify an entity of unknown origin into one of a number of available groups. Before this can be done, the unknown parameters in the logistic discriminant functions must be estimated from training data. In the case of more than two groups this requires iterative maximisation of the relevant likelihood function. Since suitable software for this purpose is not generally available, an alternative approach, called the individualised binary approach, was proposed by Begg and Gray (1984) and subsequently widely applied. In this paper the classification performance of the individualised binary approach is compared to that of the classification function obtained from maximisation of the likelihood function. The latter approach is recommended, since it is generally found to be superior. This superiority increases with a decrease in the prevalence of the reference group, with the introduction of correlation amongst the feature variables, and with an increase in the ratio of the number of feature variables to the total training sample size.

1. Introduction

Consider the problem of classifying an entity of unknown origin into one of a set of exhaustive and mutually exclusive groups, on the basis of measurements which are available on k feature variables. There are a number of techniques that can be used in this context, of which discriminant analysis, neural networks, classification trees and logistic regression are popular choices. In the literature there are numerous contributions discussing these techniques. This paper is concerned with logistic regression. The majority of the papers in the literature dealing with logistic regression focus on the case of two groups. When logistic regression is applied to classification problems in the multi-group case (i.e. the case of more than two groups), there are two approaches that can be used to estimate the parameters in the logistic regression functions, viz. a fully polychotomous (FP) approach and a so-called individualised binary (IB) approach. The IB approach was proposed by Begg and Gray (1984) as a method to approximate the maximum likelihood parameter estimates that are obtained via the FP approach. Many of the available software packages do not offer the facility of implementing a FP approach. Even when such an option nominally exists, it is often only the case of parallel regressions that is in fact offered. It has therefore become widespread practice to use the IB approach. The purpose in this paper is to compare the two approaches with respect to the classification performance of the resulting logistic regression functions. In Section 2 we briefly explain these two approaches and emphasise the differences between them. Section 3 is devoted to a discussion of a simulation study that was done to compare the classification performance associated with the two approaches. Based on the results of this comparison, we conclude in Section 4 with a number of recommendations to users of logistic regression.

2. Polychotomous logistic regression: two approaches

We require the following notation. Suppose we have $G+1$ groups, denoted by $\Pi_0, \Pi_1, \dots, \Pi_G$. Denote the prior probabilities of groups $\Pi_0, \Pi_1, \dots, \Pi_G$ by $\pi_0, \pi_1, \dots, \pi_G$, respectively, with $\sum_{i=0}^G \pi_i = 1$. An entity of unknown origin is classified on the basis of a k -vector $\mathbf{x}$ consisting of observations on k feature variables. Let $f_i(\cdot)$ be the probability density function (p.d.f.) of $\mathbf{X}$ in Π_i , $i = 0, 1, \dots, G$. Then

$$\tau_j(\mathbf{x}) = \frac{\pi_j f_j(\mathbf{x})}{\sum_{i=0}^G \pi_i f_i(\mathbf{x})}, \quad j = 0, 1, \dots, G, \quad (2.1)$$

are the posterior probabilities of group membership. The entity is classified into group Π_i if $\tau_j(\mathbf{x}) = \max \{\tau_i(\mathbf{x}), i = 0, 1, \dots, G\}$.

It is often more convenient to work with the logarithm of the ratios of the posterior probabilities. Without loss of generality, choose Π_0 as reference population, and consider

$$\begin{aligned} \zeta_{i0}(\mathbf{x}) &= \log \{\tau_i(\mathbf{x})/\tau_0(\mathbf{x})\} \\ &= \log (\pi_i/\pi_0) + \log \{f_i(\mathbf{x})/f_0(\mathbf{x})\}, \quad i = 1, \dots, G. \end{aligned} \quad (2.2)$$

Then the following classification rule, $C(\mathbf{x})$, can be formulated in terms of these log ratios:

$$C(\mathbf{x}) = \begin{cases} 0 & \text{if } \zeta_{i0}(\mathbf{x}) \leq 0 \\ j & \text{if } \zeta_{i0}(\mathbf{x}) \leq \zeta_{j0}(\mathbf{x}), \end{cases} \quad i, j = 1, 2, \dots, G; \quad i \neq j. \quad (2.3)$$

If the prior probabilities and the group conditional probability density functions were known, classification of an entity with feature vector $\mathbf{x}$ could be based on the exact values of the $\zeta_{i0}(\mathbf{x})$, $i = 1, \dots, G$. In practice the

prior probabilities are usually unknown and therefore have to be estimated. The group conditional probability density functions are also often completely unknown, or the functional form may be known, but some parameters may be unknown. In order to estimate these unknown parameters and/or density functions, it is assumed that data are available on entities with known group membership, the so-called training or learning data. The training data consist of measurements of the k feature variables on each of a total of n entities.

Several approaches to classification using (2.3) exist, depending on the degree to which parametric assumptions regarding the group conditional densities are made. Logistic regression is a partially parametric technique: no assumptions regarding the precise functional form of the group conditional probability density functions, $f_i(\mathbf{x})$, $i = 0, 1, \dots, G$, are made, but it is assumed that the logarithms of the ratios of the probability density functions are linear functions of $\mathbf{x}$, i.e.

$$\log \{f_i(\mathbf{x})/f_0(\mathbf{x})\} = \tilde{\beta}_{0i} + \beta'_{1i}\mathbf{x}, \quad i = 1, \dots, G.$$

For this model the log ratios of the posterior probabilities are given by

$$\zeta_{i0}(\mathbf{x}) = \log(\pi_i/\pi_0) + \tilde{\beta}_{0i} + \beta'_{1i}\mathbf{x} = \beta_{0i} + \beta'_{1i}\mathbf{x}, \quad i = 1, \dots, G \quad (2.4)$$

The parameters β_{0i} and β_{1i} ($i = 1, \dots, G$) have to be estimated from the training data, usually by means of maximum likelihood estimation, requiring application of an iteratively reweighted least squares algorithm. If β_{0i} and β_{1i} , $i = 1, \dots, G$, in (2.4) and (2.3) are replaced by their maximum likelihood estimates, the logistic discriminant rule is obtained.

The performance of a classification rule can be assessed in terms of its associated error rates or misclassification probabilities. The actual error rate of a discriminant rule is the probability of misclassifying a future observation,

conditional on the training data. The expectation of the actual error rate taken with respect to the distribution of the training data, is called the unconditional error rate.

The two approaches to estimation of the parameters will now be discussed for the case $G = 2$. We require the following notation with respect to the training data. Let κ_0 , κ_1 and κ_2 be subsets of $\{1, 2, \dots, n\}$ such that $i \in \kappa_j$ if and only if $\mathbf{x}_i$ is from group $\prod_j$, $j = 0, 1, 2$. Let $n_j = \#(\kappa_j)$, and $n = n_0 + n_1 + n_2$. For the FP approach, the posterior probabilities of group membership are given by the following expressions:

$$\tau_0(\mathbf{x}) = \frac{1}{1 + \sum_{i=1}^2 \exp(\beta_{0i} + \beta'_{1i}\mathbf{x})} \quad (2.5a)$$

and

$$\tau_j(\mathbf{x}) = \frac{\exp(\beta_{0j} + \beta'_{1j}\mathbf{x})}{1 + \sum_{i=1}^2 \exp(\beta_{0i} + \beta'_{1i}\mathbf{x})}, \quad j = 1, 2. \quad (2.5b)$$

The unknown parameters in (2.5a) and (2.5b) are estimated by maximising the conditional likelihood function, given by

$$\prod_{j \in \kappa_0} \tau_0(\mathbf{x}_j) \prod_{j \in \kappa_1} \tau_1(\mathbf{x}_j) \prod_{j \in \kappa_2} \tau_2(\mathbf{x}_j), \quad (2.6)$$

cf. Hosmer and Lemeshow, 1989, p.218. This requires application of an iterative procedure. It is important to note that all $2(k+1)$ unknown parameters appear in each of the three factors in (2.6).

The IB approach proposed by Begg and Gray (1984), entails performing separate binary logistic regressions to approximate the values of the estimates obtained by maximising (2.6). In the case of three groups, this is done by choosing one of the groups as reference group, say Π_0 , and performing two

separate binary logistic regression analyses involving groups Π_0 and Π_1 , and Π_0 and Π_2 respectively.

For the IB approach, the posterior probabilities of group membership are given by the following expressions. Firstly, for the binary logistic regression involving groups Π_0 and Π_1 :

$$\tilde{\tau}_{01}(\mathbf{x}) = \frac{1}{1 + \exp(\beta_{01} + \beta'_{11}\mathbf{x})} \quad \text{and} \quad \tilde{\tau}_1(\mathbf{x}) = \frac{\exp(\beta_{01} + \beta'_{11}\mathbf{x})}{1 + \exp(\beta_{01} + \beta'_{11}\mathbf{x})}, \quad (2.7)$$

and secondly, for groups Π_0 and Π_2 :

$$\tilde{\tau}_{02}(\mathbf{x}) = \frac{1}{1 + \exp(\beta_{02} + \beta'_{12}\mathbf{x})} \quad \text{and} \quad \tilde{\tau}_2(\mathbf{x}) = \frac{\exp(\beta_{02} + \beta'_{12}\mathbf{x})}{1 + \exp(\beta_{02} + \beta'_{12}\mathbf{x})}. \quad (2.8)$$

The parameters in (2.7) are estimated by maximising the conditional likelihood function

$$\prod_{j \in \kappa_0} \tilde{\tau}_{01}(\mathbf{x}_j) \prod_{j \in \kappa_1} \tilde{\tau}_1(\mathbf{x}_j), \quad (2.9)$$

while those in (2.8) are estimated by maximising the likelihood function

$$\prod_{j \in \kappa_0} \tilde{\tau}_{02}(\mathbf{x}_j) \prod_{j \in \kappa_2} \tilde{\tau}_2(\mathbf{x}_j). \quad (2.10)$$

Although the two approaches are parametrically equivalent, it is clear from (2.6), (2.9) and (2.10) that the estimates obtained from the IB approach differ in general from the estimates obtained when a FP analysis is performed. Begg and Gray (1984) studied the asymptotic efficiency of the estimates obtained from the IB approach relative to those obtained from application of the FP approach. They found that these efficiencies are generally high in the case of parameter estimation, but that "occasionally a predicted (posterior) probability will be estimated with a more substantial loss of efficiency". This occurred most frequently in cases where only a small part of the training data came from the

reference group. It is therefore not unreasonable to expect these two approaches to yield classification rules which differ with respect to error rates, especially in cases of low prevalence of the reference group. An important question that therefore deserves attention is: What price is paid in terms of classification performance if the IB approach is used instead of the FP approach, and which factors play the major role in causing such a potential loss of classification efficiency? This was investigated by means of a simulation study which is discussed in Section 3.

3. Monte Carlo simulation study

A Monte Carlo simulation study was done to compare the classification performance of the logistic discriminant rules obtained when using the two different approaches discussed in Section 2. Different underlying distributions for the feature variables $x_1, x_2, \dots, x_k$ were included in the study. Firstly, the case where the feature variables are normally distributed, was considered. Two further distributions were studied to investigate the effect of non-normality: the double exponential distribution, representing a heavy-tailed alternative to the normal distribution, and the lognormal distribution as an example of a skewed distribution.

For the purpose of this study, the actual error rates associated with the two different approaches were estimated by means of Monte Carlo simulation. To achieve this, a training data set from the relevant distribution was generated and for each of the two approaches the logistic discriminant rule was formed. The actual error rates conditional on this specific training data set were then estimated by calculating the misclassification rates when both rules were used to classify a large number (5000 per group) of entities generated independently

from the same distribution as the training data. This process was repeated 1000 times at each parameter configuration, each time generating a new training data set and estimating the actual error rates in the same way. Finally the unconditional error rate of each of the two approaches at each parameter configuration was estimated by averaging the 1000 actual error rates.

For each of the distributions mentioned above, a total of twenty-four cases was investigated. These cases correspond to specific choices of the following factors: the number, k , of feature variables, the covariance structure of these variables, the total sample size, n , and the fraction, ζ , of this total which comes from the reference group. Two values of k were used: $k = 2$ and $k = 10$. With respect to the covariance structure, two choices were made: $\Sigma = \mathbf{I}$, representing independent feature variables with unit variances, and

$$\Sigma = \begin{bmatrix} 1 & \rho & \cdots & \cdots & \rho \\ \rho & \ddots & & & \vdots \\ \vdots & & \rho & \ddots & \vdots \\ \vdots & \vdots & & \ddots & \rho \\ \rho & \rho & \cdots & \rho & 1 \end{bmatrix} \quad (3.1)$$

in which case the feature variables have unit variances and are equicorrelated. A relatively large correlation value, $\rho = 0.9$ was used. Of course, there are many other possibilities as far as specification of Σ is concerned. We investigated two total training sample sizes, viz. $n = 75$ and $n = 300$, respectively representing small and large training samples. Finally, regarding the fraction of the total sample size allocated to the reference group, we considered $\zeta = 0.2$, $\zeta = 0.33$, and $\zeta = 0.6$. In all cases the two non-reference groups were equally represented in the training sample.

The following coding will be used in the discussion below, as well as in the figures, to identify the different cases. Cases N1–N3 refer to the normal distribution, $k = 2$, $n = 75$, $\rho = 0$ and $\zeta = 0.2$, $\zeta = 0.33$ and $\zeta = 0.6$, respectively. Cases N4–N6 are similar to N1–N3, but with $\rho = 0.9$. Cases N7–N12 are similar to cases N1–N6, but with $n = 300$. Finally, cases N13–N24 are similar to N1–N12, but with $k = 10$. The same coding is used for the double exponential and lognormal distributions, but with D and L respectively replacing N .

3.1 The normal case

In this section it is assumed that the random vector $\mathbf{X}$ is multivariate normally distributed with mean vector μ_i , and covariance matrix Σ in group Π_i , $i = 0, 1, 2$. The separation between three groups can be described in terms of three Mahalanobis distances, Δ_{01} , Δ_{02} and Δ_{12} , where $\Delta_{ij}^2 = (\mu_i - \mu_j)' \Sigma^{-1} (\mu_i - \mu_j)$ for $i, j = 0, 1, 2$; $i \neq j$. There are of course many ways in which these distances can be varied. For the purpose of this study, attention was restricted to the equi-distant case, with $\Delta_{01} = \Delta_{02} = \Delta_{12} = \Delta$ (say). The following values of Δ^2 were used: $\Delta^2 = 0, 0.5, 1, 1.5, 2, 3$ and 4 . To achieve these distances in the case of uncorrelated feature variables, the elements of μ_0 , μ_1 and μ_2 , were chosen as follows:

$$\mu_{0j} = 0, \quad j = 1, \dots, k,$$

$$\mu_{11} = \Delta \text{ and } \mu_{1j} = 0, \quad j = 2, \dots, k,$$

$$\mu_{21} = \Delta/2 \text{ and } \mu_{2j} = \sqrt{3}\Delta / (2\sqrt{k-1}), \quad j = 2, \dots, k.$$

In the equicorrelated case, as in (3.1), the following choices were made:

$$\mu_{0j} = 0, \quad j = 1, \dots, k,$$

$$\mu_{11} = \Delta \text{ and } \mu_{1j} = 0, \quad j = 2, \dots, k,$$

$$\mu_{21} = a \text{ and } \mu_{2j} = b, \quad j = 2, \dots, k,$$

$$\text{with } a = \Delta/2 - (k-1)\sigma^{ij}b/\sigma^{ii}$$

$$b = \sqrt{\frac{\frac{3}{4}\Delta^2\sigma^{ii}}{(k-1)\sigma^{ij}\{1 - [(k-1)(\sigma^{ij})^2/(\sigma^{ii})^2] + [(k-2)\sigma^{ij}/\sigma^{ii}]\}}}. \quad (3.2)$$

In these equations σ^{ii} represents any diagonal element of Σ^{-1} and σ^{ij} , $i \neq j$, represents any off-diagonal element of Σ^{-1} . Data were generated from the multivariate normal distribution with mean 0 and covariance matrix Σ by means of the IMSL routine DRNMVN, and the relevant components of μ_1 and μ_2 were added to the data from groups Π_1 and Π_2 .

For each of the 24 cases identified and at each value of Δ^2 , the simulation output consists of 1000 replicates of the actual error rates of the two approaches. Averaging each of the two sets of 1000 actual error rate values provides estimates of the expected actual error rates (unconditional error rates) of the two approaches. This is the most obvious way of comparing the error rate performance. However, investigation of the actual error rate values indicates that a more detailed summary will be informative. It was therefore decided to summarise the simulation output by means of boxplots. A selection of boxplots for the normal case is shown in Figures 1 and 2. Each of the figures represents a fixed Mahalanobis distance. The boxplots were constructed using S-Plus. The notches in the boxes indicate the respective medians of the actual error rates. If

the notches do not overlap, it indicates a difference in location at a rough 5% significance level (cf. the S-PLUS Reference Manual, 1991). In addition, the means and standard deviations of the actual error rates for a selection of cases appear in Table I. In Table II, the corresponding medians and quartile deviations are given.

Since the results for the correlated cases and the corresponding uncorrelated cases are very similar, only uncorrelated cases are shown in Figures 1 and 2, and Tables I and II. Although the error rates of the two approaches decrease with an increase in Δ^2 , their relative magnitude remains the same, and therefore the discussion is restricted to $\Delta^2 = 1$. The FP approach never performs worse than the IB approach, and frequently performs significantly better. From Figures 1 and 2 and Table I, it follows that this is true in all the cases where $\zeta = 0.2$, i.e. the cases where the reference group is small relative to the other groups. In this respect, the relative error rate behaviour of the two approaches is in agreement with the findings of Begg and Gray (1984) regarding the asymptotic relative efficiency of estimation. Perusal of Figure 2 and Tables I and II shows that when $k = 10$, the FP approach also significantly outperforms the IB approach for $\zeta = 0.33$. Although the prevalence of the reference group is the main factor influencing the relative performance, it also seems that the IB approach deteriorates with an increase in the ratio of k to n . Figures 1 and 2 and Tables I and II also provide some evidence of larger variability in the actual error rates of the IB approach.

In conclusion, for the normal cases considered in our Monte Carlo study, the practitioner who adopts the IB approach, runs the risk of inferior classification performance.

3.2 The double exponential case

The double exponential distribution was included in the simulation study as an example of a heavy-tailed symmetrical distribution. For the uncorrelated case, the required double exponential observations were obtained by appropriate transformation of independent uniform random variables. For the equicorrelated case, with Σ as in (3.1), the following approach was used.

Consider a random vector $Z = [Z_1, \dots, Z_k]'$ that is multivariate normally distributed with $E(Z) = 0$ and with covariance matrix as in (3.1). Then $U_j = \Phi(Z_j)$, $j = 1, \dots, k$, are uniform(0,1) random variables. Now suppose G is some given cumulative distribution function. Then $Y_j = G^{-1}(U_j)$, $j = 1, \dots, k$, will be random variables, each with marginal distribution function G . The question now arises: given that Z has covariance matrix Σ , and that the Y_j , $j = 1, \dots, k$, are obtained from Z as described above, what can be said about the covariance matrix of $Y = [Y_1, \dots, Y_k]'$? This seems to be a difficult question to answer in general. For the double exponential distribution with mean $\mu = 0$ and variance $\sigma^2 = 1$,

$$G(t) = \begin{cases} \frac{1}{2}e^{\sqrt{2}t} & , \text{ if } t \leq 0 \\ 1 - \frac{1}{2}e^{-\sqrt{2}t} & , \text{ if } t > 0, \end{cases} \quad (3.3)$$

and therefore

$$G^{-1}(s) = \begin{cases} \frac{1}{\sqrt{2}} \log(2s) & , \text{ if } 0 < s \leq 0.5 \\ -\frac{1}{\sqrt{2}} \log(2(1-s)) & , \text{ if } 0.5 < s < 1. \end{cases} \quad (3.4)$$

Simulation experiments were conducted with this G and G^{-1} , using a number of values for ρ in (3.1). These experiments indicated that by taking $\rho = 0.905$ in the covariance matrix (3.1) of Z , a covariance matrix is obtained for the random vector Y that is very nearly equal to (3.1) with $\rho = 0.9$. Based on these findings double exponential values $X_1, \dots, X_k$ with mean $\mu = 0$ and

variance $\sigma^2 = 1$, and with covariance matrix approximately given by (3.1) with $\rho = 0.9$, were generated by taking

$$X_j = G^{-1}(\Phi(Z_j)), \quad j = 1, \dots, k,$$

where $Z_1, \dots, Z_k$ satisfy the multivariate normal requirements stated above, and with G^{-1} as in (3.4). This was done for all the groups, and the required Mahalanobis distances were then obtained by adding the appropriate values of μ_{ij} , $i = 1, 2, j = 1, \dots, k$, to the observations generated for groups Π_1 and Π_2 .

The 24 cases described in Section 3.1 for the normal distribution were also investigated for the double exponential distribution. A representative selection of the results is displayed in Figures 3 and 4 and Tables III and IV. As in the normal case, the FP approach never performs worse than the IB approach, always performs better for cases where $\zeta = 0.2$, and also performs better for cases where $k = 10$ and $\zeta = 0.33$ (see Figures 3 and 4 and Tables III and IV). The results in the double exponential case, however, differ from those in the normal case in the following important respect: the introduction of correlation adversely affects the relative performance of the IB approach. This is clear from Figure 3 for the $\zeta = 0.33$ and the $\zeta = 0.6$ cases. Once more, the IB approach cannot be recommended.

3.3 The lognormal case

To compare the classification performance of the two approaches in the case of a skewed distribution, data were generated from the multivariate lognormal distribution. The same 24 cases described in Section 3.1 for the normal case, were included in this investigation. The Johnson translation system (Johnson, 1987) was used to generate the data. A k -dimensional variable Z was

generated from the multivariate normal distribution with mean vector $\mathbf{0}$ and covariance matrix Σ , using the IMSL routine DRNMVN. The components of $\mathbf{Z}$ were then transformed as follows to yield lognormal variables:

$$X_{ij} = \lambda_{ij} \exp(Z_{ij}) + \xi_{ij}, \quad i = 0, 1, 2; \quad j = 1, \dots, k.$$

For the uncorrelated case, $\Sigma = \mathbf{I}$ was used. For the correlated case, simulation experiments similar to those described above for the double exponential distribution were conducted. From these experiments it was concluded that using $\rho = 0.935$ in (3.1) for the multivariate normal distribution results in a covariance matrix as in (3.1) for the lognormal variables with ρ approximately equal to 0.9. The shape of the resulting lognormal distribution is determined by the means and variances of the original normal variables. The parameters λ_{ij} and ξ_{ij} do not affect the shape of the distribution, but control the scale and location of the X_{ij} .

For each of the 24 cases studied, the actual error rates of the two techniques were estimated by simulation at each of the following values of Δ^2 : 0, 0.5, 1, 1.5, 2, 3 and 4. To obtain these squared distances in the case of uncorrelated feature variables, the following choices of the parameters λ_{ij} and ξ_{ij} were made:

$$\lambda_{ij} = 1/\sqrt{e^2 - e}, \quad i = 0, 1, 2; \quad j = 1, \dots, k,$$

$$\xi_{0j} = -e^{1/2}/\sqrt{e^2 - e}, \quad j = 1, \dots, k,$$

$$\xi_{11} = \Delta - e^{1/2}/\sqrt{e^2 - e}; \quad \xi_{1j} = -e^{1/2}/\sqrt{e^2 - e}, \quad j = 2, \dots, k,$$

$$\xi_{21} = \Delta/2 - e^{1/2}/\sqrt{e^2 - e};$$

$$\xi_{2j} = \sqrt{3}\Delta/(2\sqrt{k-1}) - e^{1/2}/\sqrt{e^2 - e}, \quad j = 2, \dots, k.$$

These choices yield lognormal variables with

$$\mu_{0j} = 0, \quad j = 1, \dots, k,$$

$$\mu_{11} = \Delta, \quad \mu_{1j} = 0, \quad j = 2, \dots, k,$$

$$\mu_{21} = \Delta/2, \quad \mu_{2j} = \sqrt{3}\Delta / (2\sqrt{k-1}), \quad j = 2, \dots, k,$$

$$\text{and } \sigma_{jj}^2 = 1, \quad j = 1, \dots, k.$$

For correlated feature variables, the parameters λ_{ij} and ξ_{ij} were chosen as follows:

$$\lambda_{ij} = 1/\sqrt{e^2 - e}, \quad i = 0, 1, 2; \quad j = 1, \dots, k,$$

$$\xi_{0j} = -e^{1/2}/\sqrt{e^2 - e}, \quad j = 1, \dots, k,$$

$$\xi_{11} = \Delta - e^{1/2}/\sqrt{e^2 - e}; \quad \xi_{1j} = -e^{1/2}/\sqrt{e^2 - e}, \quad j = 2, \dots, k,$$

$$\xi_{21} = a - e^{1/2}/\sqrt{e^2 - e}; \quad \xi_{2j} = b - e^{1/2}/\sqrt{e^2 - e}, \quad j = 2, \dots, k,$$

with a and b given by (3.2).

These choices yield lognormal variables with

$$\mu_{0j} = 0, \quad j = 1, \dots, k,$$

$$\mu_{11} = \Delta \text{ and } \mu_{1j} = 0, \quad j = 2, \dots, k,$$

$$\mu_{21} = a \text{ and } \mu_{2j} = b, \quad j = 2, \dots, k.$$

In the lognormal case, results are reported only for $\Delta^2 = 1$ and 2. At larger values of Δ^2 , non-existence of the maximum likelihood estimates of the parameters becomes a problem. As pointed out by Lesaffre and Albert (1989), this may be due to the populations becoming completely separated. Boxplots for a selection of the simulation output at $\Delta^2 = 1$ are given in Figures 5

and 6, with Table V containing the means and standard deviations and Table VI containing the medians and quartile deviations.

It is clear from the figures that the FP approach is appreciably better than the IB approach in all correlated cases. As before, the difference between the two approaches becomes smaller with increasing ζ . This difference vanishes to a large extent for the uncorrelated cases. A comparison of the results for the three distributions that were studied, reveals that the IB approach fares worst in the lognormal case with correlated feature variables.

4. Conclusions and recommendations

In conclusion, the FP approach is preferable to the IB approach for all three distributions that were investigated. Of the remaining factors which were varied in the simulation study, viz. the total training sample size, the number of feature variables, the degree of correlation amongst the feature variables and the prevalence of the reference group, the latter two significantly affect the relative error rate behaviour of the two approaches. The ratio of the number of feature variables to the total training sample size also affects this behaviour to a lesser degree. More specifically, the degree to which the FP approach improves upon the IB approach increases (i) with a decrease in the prevalence of the reference group, (ii) with the introduction of correlation amongst double exponential or lognormal feature variables, and (iii) with an increase in the ratio of the number of feature variables to the total training sample size. The worst scenario for the IB approach investigated in our study was the case of highly correlated lognormal feature variables with a low prevalence for the reference group. Based on our findings, we would recommend users of logistic discrimination to apply the FP approach whenever possible. It is difficult to determine which of

the standard statistical packages do employ the FP approach – documentation in the various manuals is insufficient. Examples of packages which do use the FP approach, are S-Plus and GLIM, in both cases via the generalised linear model methodology. In cases where the available software does not provide this option, or where doubt on this point exists, we recommend that the group with the highest prevalence should be used as reference group. Even then one should be aware of the fact that application of the IB approach may result in poor classification performance, especially if the feature variables are highly correlated and have a skewed distribution.

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Table I. Means and standard deviations of actual error rates, FP and IB approaches, normal data ($\rho = 0$)

	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
$\Delta^2 = 1$	FP	IB	FP	IB	FP	IB
k=2, n=75	0.487 (0.017)	0.498 (0.022)	0.465 (0.014)	0.468 (0.014)	0.543 (0.032)	0.544 (0.032)
k=10, n=75	0.531 (0.022)	0.543 (0.054)	0.521 (0.022)	0.532 (0.024)	0.557 (0.024)	0.558 (0.026)
k=2, n=300	0.477 (0.010)	0.485 (0.010)	0.456 (0.000)	0.457 (0.000)	0.537 (0.017)	0.538 (0.017)
k=10, n=300	0.491 (0.010)	0.499 (0.010)	0.475 (0.010)	0.477 (0.010)	0.539 (0.014)	0.540 (0.014)

Table II. Medians and quartile deviations of actual error rates, FP and IB approaches, normal data ($\rho = 0$)

	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
$\Delta^2 = 1$	FP	IB	FP	IB	FP	IB
k=2, n=75	0.484 (0.011)	0.495 (0.015)	0.463 (0.006)	0.465 (0.007)	0.539 (0.021)	0.541 (0.022)
k=10, n=75	0.529 (0.015)	0.548 (0.019)	0.518 (0.016)	0.530 (0.016)	0.556 (0.016)	0.557 (0.017)
k=2, n=300	0.476 (0.005)	0.484 (0.008)	0.456 (0.003)	0.457 (0.003)	0.536 (0.011)	0.537 (0.011)
k=10, n=300	0.491 (0.007)	0.499 (0.007)	0.474 (0.006)	0.476 (0.006)	0.539 (0.010)	0.540 (0.009)

Table III. Means and standard deviations of actual error rates, FP and IB approaches, double exponential data, $n=300$

$\Delta^2 = 2$	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
	FP	IB	FP	IB	FP	IB
$k=2, \rho = 0$	0.345 (0.026)	0.364 (0.042)	0.313 (0.010)	0.321 (0.026)	0.417 (0.049)	0.424 (0.051)
$k=2, \rho = 0.9$	0.344 (0.029)	0.371 (0.045)	0.310 (0.010)	0.324 (0.026)	0.406 (0.051)	0.422 (0.054)
$k=10, \rho = 0$	0.428 (0.028)	0.427 (0.093)	0.411 (0.026)	0.424 (0.056)	0.457 (0.033)	0.452 (0.051)
$k=10, \rho = 0.9$	0.398 (0.032)	0.382 (0.100)	0.383 (0.028)	0.391 (0.066)	0.433 (0.037)	0.431 (0.056)

Table IV. Medians and quartile deviations of actual error rates, FP and IB approaches, double exponential data, $n=75$

$\Delta^2 = 2$	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
	FP	IB	FP	IB	FP	IB
$k=2, \rho = 0$	0.340 (0.016)	0.355 (0.024)	0.311 (0.007)	0.315 (0.009)	0.412 (0.034)	0.417 (0.035)
$k=2, \rho = 0.9$	0.340 (0.017)	0.361 (0.028)	0.307 (0.007)	0.316 (0.012)	0.398 (0.037)	0.416 (0.040)
$k=10, \rho = 0$	0.425 (0.020)	0.450 (0.037)	0.409 (0.017)	0.427 (0.025)	0.453 (0.023)	0.458 (0.025)
$k=10, \rho = 0.9$	0.397 (0.020)	0.410 (0.058)	0.379 (0.019)	0.398 (0.029)	0.431 (0.025)	0.436 (0.030)

Table V. Means and standard deviations of actual error rates, FP and IB approaches, lognormal data, $k=2$

	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
$\Delta^2 = 1$	FP	IB	FP	IB	FP	IB
$n=75, \rho = 0$	0.303 (0.071)	0.294 (0.084)	0.332 (0.051)	0.334 (0.048)	0.550 (0.063)	0.558 (0.058)
$n=75, \rho = 0.9$	0.285 (0.105)	0.319 (0.113)	0.263 (0.078)	0.289 (0.073)	0.487 (0.099)	0.508 (0.095)
$n=300, \rho = 0$	0.286 (0.045)	0.284 (0.047)	0.321 (0.030)	0.323 (0.024)	0.574 (0.036)	0.586 (0.032)
$n=300, \rho = 0.9$	0.264 (0.081)	0.324 (0.085)	0.224 (0.044)	0.246 (0.041)	0.520 (0.067)	0.555 (0.060)

Table VI. Medians and quartile deviations of actual error rates, FP and IB approaches, lognormal data, $k=2$

	$\zeta = 0.2$		$\zeta = 0.33$		$\zeta = 0.6$	
$\Delta^2 = 1$	FP	IB	FP	IB	FP	IB
$n=75, \rho = 0$	0.275 (0.030)	0.270 (0.033)	0.322 (0.031)	0.328 (0.031)	0.558 (0.046)	0.565 (0.040)
$n=75, \rho = 0.9$	0.244 (0.071)	0.308 (0.083)	0.238 (0.044)	0.277 (0.046)	0.485 (0.080)	0.513 (0.073)
$n=300, \rho = 0$	0.272 (0.020)	0.265 (0.018)	0.316 (0.018)	0.322 (0.015)	0.576 (0.025)	0.587 (0.022)
$n=300, \rho = 0.9$	0.235 (0.054)	0.306 (0.067)	0.214 (0.022)	0.243 (0.027)	0.524 (0.053)	0.566 (0.042)

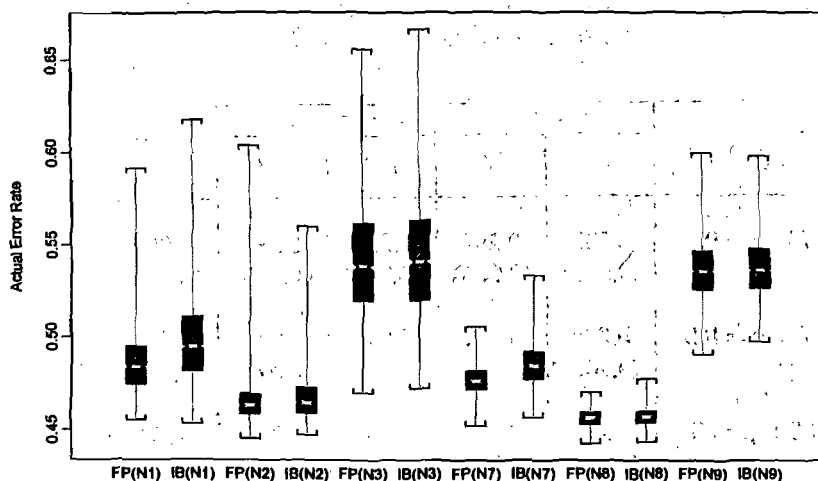


Fig. 1. Actual error rates of fully polychotomous and individualised LR normal data, two variables, correlation = 0, squared Mahalanobis distance = 1

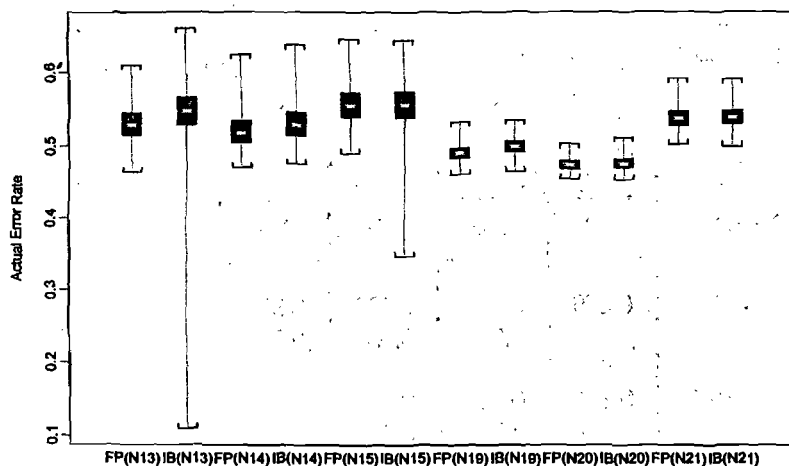


Fig. 2. Actual error rates of fully polychotomous and individualised LR normal data, ten variables, correlation = 0, squared Mahalanobis distance = 1

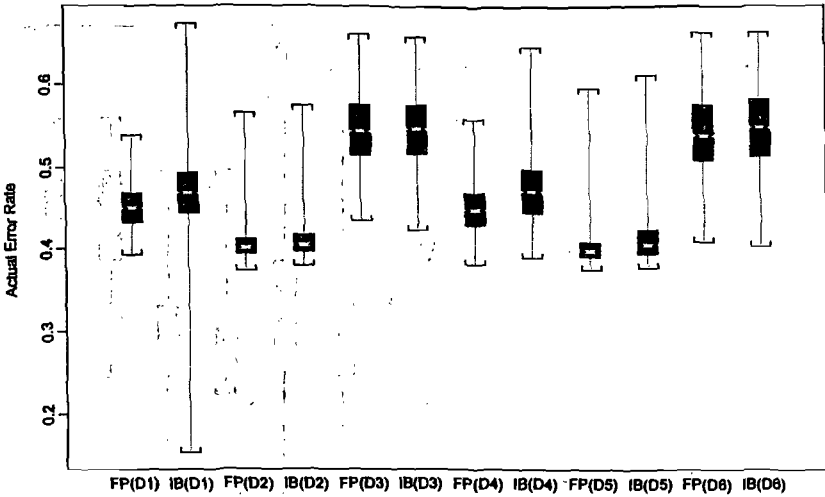


Fig. 3. Actual error rates of fully polychotomous and individualised LR double exponential data, two variables, small samples, squared Mahalanobis distance = 2

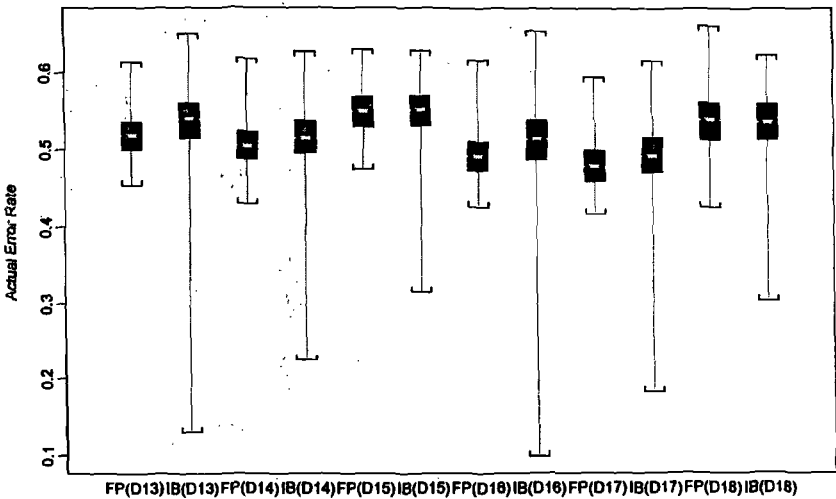


Fig. 4. Actual error rates of fully polychotomous and individualised LR double exponential data, ten variables, small samples, squared Mahalanobis distance = 2

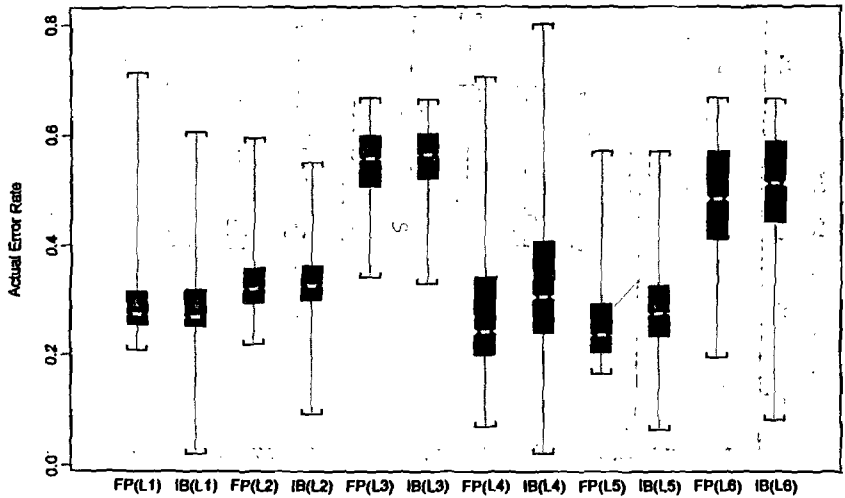


Fig. 5. Actual error rates of fully polychotomous and individualised LR lognormal data, two variables, small samples, squared Mahalanobis distance = 1

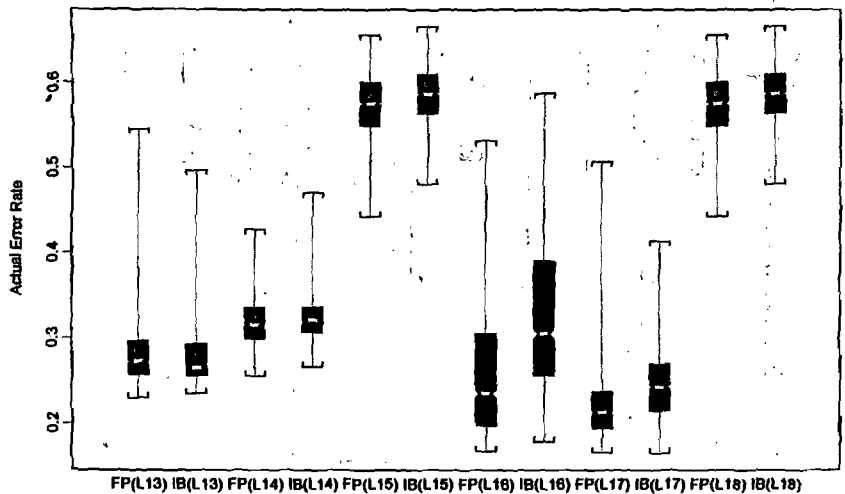


Fig. 6. Actual error rates of fully polychotomous and individualised LR lognormal data, two variables, large samples, squared Mahalanobis distance = 1